AI Analytics Agent

Complete Developer & User Guide

Written in plain English — for everyone from beginners to developers

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1. What Is This App?

Imagine you have a spreadsheet full of sales numbers, customer data, or financial records. Normally, turning that spreadsheet into meaningful charts and business insights would take hours — you would need Excel skills, a BI tool like Tableau, or a data analyst who writes code.

The AI Analytics Agent removes all of that friction. You upload your spreadsheet, add a short description of what it contains, press one button, and within seconds you get:

- A visual dashboard with charts for every important number
- Automatically detected KPIs (Key Performance Indicators) specific to your data
- AI-written insights — what is going well, what needs attention, and what to do about it
- A health score that summarises how your business is performing at a glance

No coding. No BI licence. No data analyst needed.

> □ *Think of this app as having a junior data analyst available 24/7 who reads your spreadsheet and immediately tells you what matters.*

2. The Problem It Solves

The Old Way

Most organisations sit on valuable data but struggle to act on it quickly because:

- Setting up BI tools like Power BI or Tableau takes days and requires training
- Hiring a data analyst or consultant is expensive
- Writing Python or SQL to analyse data requires technical expertise
- Pre-built dashboards only work if your data fits their rigid template

The New Way — With This App

This app is domain-agnostic, meaning it works whether your data is about retail sales, hospital patients, HR employees, or financial transactions. The AI reads your data, understands the context you provide, and automatically figures out what is important.

- Upload any CSV or Excel file — no predefined template needed
- The AI infers what each column means and which numbers matter most
- Charts, insights, and recommendations are generated automatically
- The whole process takes under 60 seconds

3. Key Features Explained

3.1 Auto KPI Detection

A KPI (Key Performance Indicator) is a number that tells you how well something important is performing — like total revenue, customer churn rate, or average order value.

Normally, someone decides in advance which KPIs to track. This app does it automatically by:

- Scanning every column in your data to identify which ones are measurable metrics
- Using AI to rank them by how important they are likely to be for your domain
- Assigning each KPI a priority (High, Medium, or Low) so you know where to focus first

> □ *Even if you never heard the term KPI before, the app will surface the numbers that matter most in plain language.*

3.2 AI-Powered Insights

After detecting KPIs, the app asks the AI to analyse them and produce four types of insight:

Insight Type	What It Means
Positives	Things that are performing well — metrics you should celebrate or maintain
Concerns	Problems or warning signs that need your attention, ranked by urgency
Trends	Whether key numbers are going up, going down, or staying stable over time
Recommendations	Specific actions to take, with the reason why and the expected benefit

3.3 Health Score (0–100)

The health score is a single number from 0 to 100 that summarises the overall state of your data:

- 75 and above — Healthy (shown in green)
- 50 to 74 — Needs Attention (shown in amber)
- Below 50 — Critical (shown in red)

It is computed by the AI based on the balance of positive signals, concerns, trend directions, and anomalies found in your data.

3.4 Flexible Metadata

Metadata is simply a description of your data — what it represents, what time period it covers, and what the columns mean. The app gives you three ways to provide this:

- Fill in a form — type a short description and optionally name your KPIs
- Upload a JSON file — if you already have a structured metadata file, just upload it
- Skip it — the AI will read the column names and sample rows and infer the context automatically

> □ *The more context you give, the more accurate and business-relevant the insights will be. But the app works even with zero metadata.*

3.5 Interactive Charts

Every KPI gets its own chart. The app automatically picks the right chart type:

- Bar charts for comparisons across categories (e.g. revenue by product)
- Line charts for trends over time (e.g. monthly orders)
- Pie charts for proportions (e.g. customer segment breakdown)
- Heatmaps showing how columns correlate with each other
- Distribution charts showing the spread of every numeric column

3.6 Data Explorer Tab

Beyond the automated analysis, you can also explore your data manually. The Data Explorer tab lets you:

- Filter which columns to show
- Scroll through your cleaned data
- See statistical summaries (averages, min, max, etc.) for every column
- Download the cleaned version of your data as a CSV file

4. How to Use the App — Step by Step

Step 1 — Open the App

Visit the deployed URL in any web browser. The app loads a landing page with a sidebar on the left.

Step 2 — Enter Your API Key (or It May Already Be Set)

In the deployed public version, the Groq API key is pre-configured — you can skip this step. If you are running the app locally, paste your free Groq key (from console.groq.com) into the API Key field.

> ⚠ *Never share your API key publicly. In the deployed version, the key is hidden server-side.*

Step 3 — Upload Your Data

Click the file uploader and select a CSV or Excel file (up to 200 MB). Excel files with multiple sheets are supported — the app will process every sheet automatically.

Step 4 — Add Metadata

Choose one of the three options:

- Fill in the form with a dataset description, domain, and time period
- Upload a JSON metadata file if you have one
- Do nothing — the AI will infer it

Step 5 — Run Analysis

Click the Run Analysis button. A progress bar shows the four stages: preprocessing, metadata inference, KPI detection, and insight generation. The whole process typically completes in 10–30 seconds.

Step 6 — Explore Your Dashboard

The dashboard opens automatically with five tabs:

- Overview — health score, executive summary, KPI metrics, distribution charts, correlation heatmap
- KPIs — each detected KPI with its chart, priority badge, and description
- Insights — positives, concerns, trends, anomalies, and recommendations
- Data Explorer — interactive table view and statistical summary
- Raw Details — the underlying JSON for metadata, KPIs, and insights

5. Advantages

5.1 From a User Perspective

Advantage	Why It Matters to You
No technical skills needed	Anyone who can use a website can generate a full analytics report
Works with any domain	Sales, HR, finance, healthcare, marketing — the AI adapts to your data
Results in under 60 seconds	No waiting days for a report from an analyst or consultant
Actionable recommendations	Not just numbers — specific things you can do to improve performance
Free to use	Groq provides fast LLM inference with a generous free tier
Data stays private	Only statistical summaries are sent to the AI — not your raw data rows

5.2 From a Developer / Technical Perspective

Advantage	Technical Benefit
Modular architecture	Each concern (ingestion, KPI, insights, charts) lives in its own module — easy to extend or swap
Structured LLM outputs	All AI calls return JSON — no brittle string parsing or hallucinated formatting
Minimal LLM calls (2–3 per run)	Keeps latency low and API costs negligible
Stateless modules	All modules are pure functions — easy to unit test in isolation
Provider-agnostic LLM client	Swapping Groq for OpenAI, Anthropic, or an open-source model requires only changing groq_client.py

Zero-config deployment	One GitHub push deploys to Streamlit Cloud with secrets managed server-side

6. Using Open-Source LLMs (No API Key Needed)

The app's LLM client is designed to be swappable. If you want to run the app entirely locally without any cloud dependency, you can replace Groq with an open-source LLM running via Ollama.

6.1 What Is Ollama?

Ollama is a free tool that lets you run large language models on your own computer. Once installed, it exposes an OpenAI-compatible API at localhost:11434 — meaning the app can talk to it without any API key.

6.2 Recommended Open-Source Models

Model	Size	Best For
llama3.1:8b	5 GB RAM	Fast responses, good JSON output, recommended starting point
mistral:7b	4 GB RAM	Strong instruction following, very fast
llama3.1:70b	40 GB RAM	Best quality, requires high-end GPU or Apple M-series Mac
phi3:mini	2 GB RAM	Ultra-lightweight, works on most laptops

6.3 How to Switch to Ollama

- Install Ollama from ollama.com and run: ollama pull llama3.1:8b
- In src/llm/groq_client.py, change the base URL to http://localhost:11434/v1
- Remove the API key validation — Ollama does not require one
- Change the model name to the Ollama model tag (e.g. llama3.1:8b)

> □ *For local development and privacy-sensitive data, Ollama is the ideal choice. Your data never leaves your machine.*

7. Industry Use Cases

Retail & E-Commerce

- Upload monthly sales CSV → instantly see top-selling products, low-margin categories, and return rate trends
- Recommendation: which product categories to promote and which to discontinue

Human Resources

- Upload employee data → detect attrition risk, salary distribution anomalies, and department headcount trends
- Recommendation: which teams are understaffed or overpaying vs. market rates

Healthcare

- Upload patient visit records → identify high-readmission diagnosis codes, average treatment costs, and seasonal demand spikes
- Recommendation: staffing adjustments and cost-reduction opportunities

Finance & Accounting

- Upload transaction ledger → auto-detect expense categories, profit margins, and cash flow trends
- Recommendation: budget reallocation and cost-cutting priorities

Marketing

- Upload campaign performance data → surface best-performing channels, cost-per-lead, and conversion rate trends
- Recommendation: budget shifts between channels based on ROI

8. How the App Was Built — Developer Notes

8.1 Project Structure

The project follows a clean separation of concerns:

- app.py — the single entry point; handles UI layout, session state, and the analysis pipeline
- src/data/ — ingestion and preprocessing; pure pandas functions with no UI dependencies
- src/metadata/ — metadata loading, validation, and LLM inference
- src/kpi/ — heuristic scanning and LLM-based KPI ranking
- src/insights/ — prompt construction and insight JSON parsing
- src/dashboard/ — all chart-rendering functions (stateless, return Plotly figures)
- src/llm/ — the thin LLM client wrapper

8.2 Why Streamlit?

Streamlit was chosen because it turns Python scripts into web apps with minimal boilerplate. Key advantages for this project:

- Reactive rerun model — any widget interaction triggers a full script rerun, keeping state consistent
- Built-in file uploader, progress bars, tabs, and metrics — no frontend HTML/CSS required
- One-click deployment to Streamlit Cloud with GitHub integration

- st.session_state provides a simple key-value store across reruns

8.3 Why Groq + Llama?

Groq's inference hardware (Language Processing Units) delivers token speeds 10–20x faster than standard GPU inference. For this app, that means:

- The entire 2–3 LLM call sequence completes in 3–8 seconds
- Users do not perceive the AI step as a bottleneck
- The free tier is sufficient for dozens of concurrent users

8.4 Prompt Engineering Approach

Each LLM call uses a system prompt that:

- Defines the exact JSON schema to return
- Provides domain context from the user's metadata
- Includes statistical summaries (not raw data) to stay within token limits
- Instructs the model to respond with only valid JSON — no preamble, no markdown fences

> ⚠ *If you change LLM providers, test the JSON output structure carefully. Different models interpret the schema instructions with varying fidelity.*

9. Deployment to Streamlit Cloud — Step by Step

Deploying takes less than 10 minutes:

- Push your project to a public or private GitHub repository
- Go to share.streamlit.io and sign in with GitHub
- Click New App, select your repository and branch, set app.py as the main file
- Open Advanced Settings and add your secret: GROQ_API_KEY = 'gsk_...'
- Click Deploy — Streamlit installs dependencies and starts the app automatically
- Share the generated URL with anyone — no login or installation needed

To use the pre-configured API key in the app, update groq_client.py to fall back to st.secrets:

- api_key = api_key or st.secrets.get('GROQ_API_KEY', '')
- This way the key field in the sidebar becomes optional for end users

10. Frequently Asked Questions

Is my data secure?

The app sends only statistical summaries (averages, counts, column names) to the AI — never your actual row data. Uploaded files exist only in your browser session and are not stored anywhere permanently.

What file formats are supported?

CSV (.csv), Excel (.xlsx, .xls). Excel files with multiple sheets are fully supported.

How large a file can I upload?

Up to 200 MB via the Streamlit file uploader. For very large files (millions of rows), the app will sample data for the AI analysis while still showing full statistics.

What if the AI gets something wrong?

AI-generated insights are starting points, not final answers. Use the Raw Details tab to inspect exactly what the model returned. You can re-run the analysis or adjust your metadata description to guide the model toward better results.

Can I use this with my company's private data?

Yes, but consider running the app locally with Ollama (see Section 6) so that no data leaves your network. For cloud deployment with sensitive data, ensure your Streamlit Cloud account is on a private plan.